

Active InferAnt Stream 006.1 ~ Towards (Open Source, Active Inference) B

ActiveInferAntStream | Towards (Open Source, Active Inference) Bioregional Biofirm Modeling | 1:50:52

All right. Hello, everyone. It's November 3rd, 2024. I'm going to begin active inference stream number 6.1 as per usual with a GitHub push, and then let's jump right into the repo, and this should be an awesome stream. So thanks to those who are joining live. For those who are interested in bioregional modeling, open source active inference toolkits, and more, this should be quite exciting. So I'm excited to walk through where things have gotten over the last few days and talk about how we'll continue over the next 10 days plus plus plus. All of this is happening in the open source repo biofirm. I'm sure we'll return to this in a lot of different ways in the coming stream, and everything that I'm about to go through is the current state on biofirm. We just push the update. Let's reload to confirm. Awesome. Welcome, John. All right. There's two aspects of the biofirm concept that the current state of the repo and this stream are going to demonstrate. The first is the active inference component, which is an active inference-based multi-agent control framework. One agent deployed per ecological modality with differing levels of noise and controllability within a general homeostatic framework. All of this, of course, work in progress, simple where possible. However, I ran into a lot of fun ways to work with PyMDP over the last few days years. Using cursor 0.42. The second part of the biofirm research is going to reuse some code from Gen24. This is going to use perplexity LLM with internet access calls with an API to do some bioregional research and structuring and reporting for California counties. Okay, where are we going? And then let's jump first into the active inference component, stream.

Okay. The goals of the stream are to first demonstrate a proof of concept for bioregional modeling using active inference frameworks, building on what Andrew Pachea and others have done. Second, to prepare, work towards preparing some open source bioregional templating for the coming applied active inference symposium. And then third, explore PyMDP methods and simulations and get a better sense for what's really happening with that ABCD after all. Okay. Detail agenda. Just to read the highlights, then we'll jump into one. One, bioregional modeling outputs and impact. Two, more foundational background. Three, we'll go into detail on PyMDP, partially observable Markov decision process, POMDP. Four, talk a little bit more about bioregional modeling, and then there's a bunch of discussion topics. All right. Here's where we're going with this simulation. So I'll walk through the visualizations first. Here is 10 ecological variables. This uses in the utils file. I'm going to jump around a little bit. So ask any questions in the chat and also look at it synchronously on your own time. ecosystemconfig.json is going to describe the variables as well as their relationships amongst them. I kept that part empty, but it does work. Each variable has an initial value, which is set at 50, and a constraint homeostatic range. Here for all the variables, it's set at 45 to 55. Then each variable has several important factors that determine its controllability. First is a trend coefficient. This does work. However, I've left them as zero for now. And then two main factors that we're going to be exploring a lot, which is the control strength and the noise standard deviation.

So what's going to happen in ecosystem simulation is it is going to run a random setting and watch the ecosystem unroll with no control. And then it's also going to run an active inference simulation where depending on the controllability, the active inference agent is able to reenter those scaled controls back into the ecosystem. So to get started, first you run from the scripts folder for now, Python 3, and then startup.py. Startup.py is going to install into a Python virtual environment, the PyMDP package. And then you need to run source, VMV bin activate in order to get it going. You'll know that you're in it because it'll say VM and it won't be like, where's PyMDP? I already downloaded that. It's in VM here. So we can skip the Python 3 startup. The big script to run is ecosystem simulation. So here's what happens when it runs. First, it brings in a bunch of dependencies. Let's run it so that we see what it looks like first, and then we'll walk through the logic. It logs variables. This is a deep debugging error, so that's not done yet. It does the active inference simulation, does the random simulation, then the active inference simulation, saves all the logging into the output folder. It's going to talk about generating these visualizations, and then we're going to go to the visualizations folder, and we're going to look at those. So in this case, while it's logging, we'll start to look through. It's going to be doing 1,000 time steps, so it'll run the same random seed for random uncontrolled setting and for active inference partial control setting for 1,000 time steps, discrete time, discrete state space, and print an extended log to the terminal every 100 time steps. It's going to inject a control action every time step. That could be changed and have some logging information, which was really a lot of the fun. Okay, here's some dictionary methods. Here's the primary environment, possibly in the future could be improved just like everything else, so many different ways, RxInfer, RxEnvironments, etc. But this is just an environment class. Loading and validating the configs reads in the number of ecological variables. So to change the number of ecological variables, this is a place where cursor does amazing things. So we're in the ecosystem config and say write 10 more ecologically relevant variables for California. Okay, meanwhile, we finished the main visualizations from the ecosystem simulation. We're going to look at them in a second. And then now it's going into this second level analysis for just the free energy principle active inference type analysis, which we'll look at second. That's not totally working. I still just wanted to do this. Now it gives enough of a direction. But just to pull back to cursor and how we can use the semantics of natural language to modify structured data. So here's that ecosystem config. And then it's added a bunch of more ecological modalities. And then if we were to run ecosystem simulation again, it would use those new modalities and make an even longer plot. So I'm just going to reject that. But that's how that works. Okay. So again, it loads in the config, including the variable relationships, which are not utilized here.

So checks for circular dependencies. Some of this is just to make a script like flow, of course, in Rx and for and so on, there's more advanced ways to deal with this type of graphical specification. Running simulation. It runs it with an active inference agent if needed. You can also call it in random only mode. What it's going to do is for each of the controllable modalities in the ecosystem config file. So each of these modalities controllable is true. If it's controllable equals false, it's something that's not going to be analyzed as controllable. Then that doesn't need to spin up a control active agent. Maybe there could be a sense making agent that gets spun up, but it's just everything's being modeled as controllable to different extents.

It runs the simulation. We're going to look more into the logic of the bio firm agent and also the POMDP ABCD matrices. It runs the simulation.

What happens at each time step of the simulation is that there's the natural underlying dynamics of the natural resource or proxy on the 0 to 100 scale continuous value. Then at each time step, that gets distilled to a 0, 1, or 2, just three discrete values, either low, within homeostasis, or high.

There's three options and that simplifies the input space for the active agent. We'll talk more about this when we get to the generative model. Then the active agent per that bio modality does hidden state inference, then policy inference, samples an action, and then returns that action as either a negative 1, which is decrease. So here it is with the observation being converted to 0, 1, or 2.

Then the active agent is going to do inference on sense making and action selection, returning negative 1, decrease, 0, maintain, or plus 1, increase. That control signal is going to get scaled by the controllability of the ecological factor. So there's the natural noise variation happening every time step, and then also if it's controllable with active inference in that case, it re-injects the control signal multiplied by the controllability. Here's the step function. Also, just to note that the

Claude 3.5 Sonnet 2024-10-22 that came out 10 or so days ago, it's markedly better with cursor 0.42 than prior. These scripts are getting to around 1,000 to 2,000 lines each, which is incredible because from even prior active inference streams, you'll remember where we were breaking things up after about 200 or 300.

Performance metrics are logged. Various logging, it's hard to know if you're going too hard or not hard enough on logging. Tracking is initiated, states are recorded. Just getting through all of this. Here's the

initialization of the active inference agent. We'll talk about how that's specified when we get to its logic after looking at the outputs. And then that's the main script. Okay, so here's what those visualizations get you and how this is just schematic. It's not meant to actually be like,

here's the kind of situations where it works or not. It's just comparing it against a simple maintain random noise strategy. We can add other strategies and other algorithms for policy selection to compare better. Okay, control actions analysis. This plots red, random for green, active inference, and it plots the occupancy. Also, each of the ecological variables, of course, they're just meant to be schematic. They're appended with the noise and the controllability. So you can start to explore, and we'll talk more about that in the parameter sweep, what is the controllability of different noise variables and controllability intensities. So looking at soil pH, which has a every time step and noise variability of 0.07, standard deviation, sampling from normal there, and then a controllability of 2.8,

which means that every time the negative one is output decrease, it goes down by 2.8 points overall. So this is quite controllable, and we see it spending a lot more of its time near the homeostatic range because it's able to control an amount that is slightly larger than the noise factor. Here's another situation with the nutrient levels where we see control 1.8 versus noise 0.6, and so it's able to navigate towards these homeostatic bands. So this is the occupancy with the control actions across these. Soil compaction, this is another interesting one. Even with a smaller control, it's able to stay within the space. Because also, as with all Gaussian noise processes, the expected value going through time of random noise is the same value as the starting value, it's just that the variance is higher.

Okay, correlation heat maps. Here's looking at the correlations amongst these variables in the random case on the left and the active inference case on the right. And we can look at certain things like soil structure and see that, okay, it's lighter here, it's darker here. So this is just to show you can look at the correlation structure of the dynamics of different variables, and whatever it is that the active policy is doing, it is changing the correlation structure.

So it's not just recapitulating random. Satisfaction rates. So satisfaction rates are the percentage of time that across all variables, each variable spends in the homeostatic zone. So it starts out high because the initial value starts at 50, which is right in the middle of their zone for all of them. And then random control diffuses out, and we can run a longer time step simulation soon. Let's actually just run it in the background to show that works, and then it will like reset. So I'm going to go to the ecosystem simulation.

Change the number of time steps to 10,000. Save it, clear it, run a simulation. First, it's going to, this is the random, so that's faster. And at each of those 100 logged time steps, it's basically oscillating like 0 to 20 percent, and there's no control effort because it's not selecting policy. And then it's giving a play-by-play, and then you'll see on the bottom that it will switch to different logging mode once the actinf comes into play. Okay, continuing on through here. So we looked at satisfaction rate. Here's another way to look at state distributions.

So this is the top is the random control, bottom is the active inference. So we can see, okay, well, in the random setting, again, these are just schematic. There's not any ecological validity.

Soil structure spent about half the time in low and half in high, and over here it spent more time in high and homeostatic. State transitions. This is looking at the history of policy selection in terms of how did the transitions amongst the different states go. So in the soil organic matter, low always went to low, almost always 0.99, and little time was spent in high at all, whereas those numbers are different. There's just one visualization that can happen. Here's the new visualizations, and they'll overwrite these. Okay, this is a great way to look at the difference between these two. Remember from the same random seed and the same underlying randomization dynamics

is happening for both active inference and for the random case. Okay, it's kind of a large image. It's easier to see some of these things when you only have like one or three modalities.

So let's...

Okay.

Here we go. So this is strategy comparison. ActInf is on the right. Natural evolution of that. Uncontrolled dynamics are on the left.

So here we can see...

They're varying in the amount of controllability and noise, but we can look at how they move in and out of that band.

System stability analysis. This is another fun one. This is just three random variables chosen, but we can see from this

phase space excursion plot, is it popping up at you? Is it pushing back away from you?

We can see that the ActInf diffusion is constrained because there's this partial homeostatic control, and there's different control theoretic metrics we can make.

Average control effort, I think this we can look at how it's calculated, but it just relates to how much change is happening step by step, and the variable distributions.

So here we have the actual active inference agent.

So here again we have that soil pH, the blue line for active inference spending much more of its time in the homeostatic range, while the gray random policy maintain plus random noise has a more diffuse shape.

So sometimes it's bimodal, and in the large limit, you can, with many more time steps, I've run several for like 50 or 100,000,

and those have more of a convergence property. So these are the kinds of things that we can study like in the coming 10 days and beyond. Let's look at the satisfaction rate.

So here it only did slightly better. So again, it's not intended to say that active inference does literally any better than random. In any given case, it doesn't necessarily do that.

It may not do better or simpler than any other kind of policy selection at all.

It's just about bringing this onto the table and looking at how PyMDP facilitates that.

And then last piece of the outputs. Each of the outputs has a modality specific folder, and these have several visualizations. Some of them are going to glitch, but it looks like this is a belief distribution, earlier versions, ancient, Ozymandias-like, earlier versions of the script had a lot more informative, super cool looking things happening here. But this is just, for now, it's the margin doesn't allow what needs to be shown in these dashboards. But there's a lot of methods and partially broken slash partially functional code. And that's what's happening now. So when you run the ecosystem main simulator, it'll run the number of time steps for random for active inference. It will do those overall high level visualizations that we just looked at in that folder. And then in the active inference analysis, it looks more like the expected free energy, belief dynamics, a bunch of other stuff. Okay. So we saw just from looking at that, that there were certain combinations that seemed to do well. And then there were certain combinations where the control was not working very well. So here, again, in an amazing example of how knowing that a certain kind of analysis would be preferable, having seen it in other papers, just thinking about it yourself and so on. Here is going to be a sweep run across control abilities. So here,

it's multi-threaded. We can use this when we can, this is super fun. I was going pretty high, given everything I have on these older computers. Here we have number of CPUs. This is going to be separate replicate virtual threads or physical threads, if you have them, for running this sweep.

Noise range and control range are going to be the log space variability. So here, let's go from 0.1 to 1. 1. And so that'll be sampling from a 10th to 1. Let's do it to 2, just like it had. And we'll do 10 points along each. So this is going to do 100 simulations across all combinations of noise and controllability. Here we'll do 200 act-infimesteps. And then this is fun, number of repeats. So here, we can run five replicates at the exact same setup to get convergence properties and stationarity at the within experimental replicate, like within quote biological replicate. But this is like doing technical replication. So again, all this is going to be useful as we continue to develop this analysis. Let's just do it with these numbers. 200 may not be enough to show differences. Let's just, but it's going to take three times 200 for each of a hundred, but it's going to divide that by eight because it's going to have eight threads on this computer. So Python 3, noise control. Okay. Okay, great. That's not taking too long. And we haven't yet looked at what is the biofirm agent itself. These are just the outputs of it and the sweeping of it. After showing this sweep analysis, I'm going to pull back to the biofirm agent and how the ABCD matrices are scaled. Okay. So here we have in outputs, sweep. This was a, must have been earlier,

set the earlier analysis. So let's look at this while it's running.

Belief entropy surface doesn't look to be calculated properly, but that's kind of cool.

Control effort surface. Again, not necessarily calculated, but fun. Satisfaction rate surface.

Whoa. So here we have...

As you have lower variability, you... and it's a short simulation. So with low variability like 0.01, even with 100 or 300 in a row, it doesn't escape the satisfaction range of plus or minus five.

Whereas we see as we go down towards higher noise, we end up spending more time outside of the homeostatic

bound. Interaction. Again, this is all just meant to be speculative realism coding.

Different control efforts.

Here we see satisfaction rates.

And see, okay, it looks like you have more control

this bottom row. When the noise is in that quintile, then this is a more control efficacy.

We can see satisfaction rate dropping as variability is higher.

Here's a heat map.

Satisfaction rate. So that's in yellow down to blue is the satisfaction rate.

Noise SD on the left, control strength on the right.

So these are probably not sampled from stationarity because it was only a few hundred time steps, but I did some runs for a few hours where

you can run 10 replicates per cell. So it gives you a strong differentiating signal at each cell

and then run each of those replicates, technical replicates for like 10,000 so that it comes to a

stationarity or use a burn burn in period. And then that starts to reveal some really interesting patterns. Okay. So there's other parameters of sweep over. I mean, I'm sure it's going to be a theme we return to again and again and again and again, which is that while the active models can be incredibly interpretable, simple, effective, et cetera, et cetera, often to understand what is really being needed. Are we trying to do sense-making on whether we're out of homeostasis or not? Because otherwise we could say observation confidence is a hundred. We just, we know a hundred percent whether we're within or low or high, or do we only want to have 55% confidence in the signal coming from the low homeostatic high. But then to the extent that actually that signal's valid, that's diluting the efficacy of the control. Or we may set a very high value for homeostasis to see more of a pullback attractor effect. But again, if it was as simple as saying low means turn on the heater and high means turn on the cooler, if it was as simple as that kind of a rule-based system, you wouldn't need to do expected free energy, pragmatic value plus epistemic value at all.

So we're going to return to this. Okay. Now let's get to what is the BioFirm agent?

All right. The act, the BioFirm agent, and again, this is drawing on Klippinger's work and also Andrew Pachea's work and how he set this up. It's an active inference agent bundle for ecosystem control. It creates N independent POMDP, partially observable Markov decision process agents, one per controllable modality. Each POMDP agent takes in a discrete observation, low, homeostatic, or high. And this is kind of fun as we talk towards what could be a generalized act-inf agent or kind of a minimal essential kernel and all that by saying, well, we're going to let the simulation map, the generative process, the environment map from whatever the value is actually out there, like in degrees to just this homeostatic variable. Or I thought, well, we could have more nuance. We could have a five-state model, like too low, getting low, just right, a little bit warm, too warm. And then that'd be a five by five. But here it's a three-state observation. Then there's three hidden states as well. Are we actually low, homeostatic, or high? That's mapped by the A matrix. The B matrix goes from two hidden state transitions with each slice being in affordance, one action policy, which is either decrease, maintain, do nothing, or stay. C is the same shape and size as a distribution over observations. And then D is just to get the chain rolling, the initial hidden state prior, which is just uniform. So what's going to happen using PiMDP is first, an observation is going to come in from the ecosystem simulation. That's going to be a one-hot, just a 1-0-0 if it's low, or 0-1-0 if it's in homeostatic, or 0-0-1 if it's high, and so on. That is going to get pushed through the A matrix, the three by three, to end up with the belief distribution over latent states. Like, is it really cold or homostatic or hot? Given the observation, that's the initial variational free energy sense-making. Then expected free energy is used for policy selection, which is, given my updated beliefs about latent state distributions, what would be the payout for each of these slices of B evaluated in terms of epistemic and pragmatic value? So here's what's going to happen. The biofirm wrapper starts by initializing the number of POMDP agents that it needs, which is the number of controllable variables that are in the situation.

There's two free variables, and again, this just depends how generalized you want. Do you just want to say it's 0.9 and 0.1 confidence, and we'll leave it at that? Or do you want to do parameter fitting on even higher order variables, learning, and so on? Observation confidence is going to be the on-diagonal element of A. We'll look at that in the POMDP. Homeostatic preference is going to be the on-homeostasis preference enrichment in C, and we'll look at that too. This was the main roadblocking and speed bumping and gapping and laddering was in and out of numpy arrays, list, object, all these different python shapes, and it's going to be awesome. Already, cursor demonstrably can make some of this happen, and building up really explicit, really good documentation. Then, even if it's online, you can bring in the POMDP documentation online. So this is super interesting. It allows us to say things in different places and have that influence the model. Like if it's like, yeah, it should be a numpy array here, and here's the method you use somewhere else. Here's an example of where it works. Use this code. It works. Then that will have really cascading effects.

The ABCD matrices are made. We're going to look at those after walking through the logic of the Bioform agent, but each controllable modality has its own agent, so it isn't using cross-latent state information. Each of these is meant to be a standalone controller. Here, we make each of those modalities have their own ABCD and check, check, check, check, check, all these different features. There might be some place where I'm doing it redundantly right now, moving it in and out of a different format, and I look forward to ironing those things out, but it works now.

So, get action

is going to take in the action

that's selected, which is again like so many of these other small things. The actions, like the affordances, are listed in terms of a zero index list, affordance zero, one, or two, whereas we want it to be into a control signal like negative one, zero, or one, kind of reverse, neutral, or forward. Then, here's the scaling by control strength. So, here in ecosystem config, we can see, okay, the controllability of microbial activity is 1.2. So, that means a positive signal for there is going to result in 1.2 efficacy on the next time step. Getting agent data, this is about logging information from the agent. So, it's actually pretty simple what's happening. However, there's a lot that's happening underneath the hood with a few different things. First, the matrices being created in this script we're going to look at in a second, and then also the PyMDP methods, which is brought in. The script clone, PyMDP clones the whole repo, which is useful again for cursor to bring into the context. So, it has all these positive examples. So, for open source inference, adding well-documented effective code of any kind, probably, is having big impacts in how these language models, through their training and context, are able to make active inference models for dreamlike scenarios. So, that's what was used during the creation of this package. I just didn't clone it onto this other computer. Agent.inferStates is the perceptual inference component, and then sampleAction is also used.

It might be missing one line here, or I can explore where else, if it's doing the expected free energy based methods. I tried several different ways for doing that EFE step, ranging from using the PyMDP methods to calculating the epistemic value and the pragmatic value separately.

Again, this is just intended to be a sort of a scaffold, and I look forward to ironing out a few of the low-hanging fruit in the coming days. Here's the POMVP matrices that are invoked for each of the modality-specific agents by the biofirm. Observation confidence, again, is going to be the on-diagonal A matrix. Homeostatic preference is going to be the middle homeostatic preference in C.

POMVP matrices is a method that wraps around specifying ABCD parameterized using the config parameters.

Checks the shape, moves a minute out of a bunch of different things.

Here's the initialization of A. So, this creates a simple 3x3 likelihood matrix mapping empirical observations to latent states. So, from a one-hot observation to a belief over hidden states, if you have confidence equals 1, 100% confidence, then it's just an identity matrix, and the one-hot observation maps to a one-hot belief over latent states. If you had confidence being higher so that noise was a non-zero value, then A has an off-diagonal component. So, what happens is first, the diagonals are filled with the confidence, then the off-diagonals are filled with these kinds of confusions, and then the other half of the noise per column is added to the mega confusions, which would be the probability of thinking that it's hot given that the observation came in cold. So, here's what it kind of ends up looking like at the end. We have FROM and TO, and the bigger numbers are on the identity axis, and then we have these noise over 2 remainder from 1 on these off-diagonals.

Another place where better usage of POMVP methods and more examples could help is in the normalizing of different A and B matrices. Here's where we get to the B matrix.

B matrix is regarding latent states. You can look at textbook group, figure 4.3 or 7.3, as we often do. B is going to intervene in how hidden states change through time. B is a tensor. Each slice of B is FROM and TO, latent states, as a matrix. And then each slice, like an index card, is which affordance is being selected. So, here's the B matrix. Policy selection is selecting out a slice of the index card of B, and then that is used as a transition operator on hidden states. So, this is broken out, and again, there could be a lot of interesting ways to improve how this is written. But here are the three actions. Decrease, maintain, and increase. And again, this is where it comes to a learning POMVP experience and for future software packages, where each of these values have a real interpretability. And here, there's no learning or updating. So, this speaks to part of the interpretability of how policies are determined in active inference is we can say this is the fixed beliefs or these are the updatable beliefs and how they're updatable about the effects of policy. So, it's not just training the weights, continuous values for n billion nodes. It's doing matrix operations with these investigatable matrices. So, here, this is the situation where we take a decrease action. So, this is kind of like turning on the air conditioning for a room. This is from and to. So, this is the probability of staying in low. Given that you're in low and you chose decrease, 90% of the time, you stay in low. 10% of the time, you go to homeostatic. 0% of the time, you go to high. Let's just look at another one. Given that we chose maintain and we started in homeostasis, our belief is that 80% of the time, we transition to stay in homeostasis and 10% of the time, we go to either hot or cold. And again, one more for increasing. It's the mirror here. It doesn't have

to be, but it is for decrease and low. I'm saying, given that we chose increase this slice of B and that we were in high, we're going to stay in high with a 90% probability, 10% going to homeostasis. So, again, this was a big thing in the fitting was this is a relatively sharp belief about the efficacy of policy. I mean, thinking that we have too few soil microbes, we took this one policy and then does it have a 90% probability of moving us to homeostasis at the next time step? Probably not. So, how do you deal with multi-scale policy nesting, different efficacies, learning on B, parameterizing B based upon causal relationships and other structures? This is just intended as a scheme. That's B. So, we looked at A, sense-making matrix, B, conditional action transition operations, now C. So, C is three elements in a list array. The low and high are given a value of 0.1, kind of arbitrarily, and then homeostatic preference is the middle value. So, if the value was four, it'd be 0.14.1. And that's reflecting the preference that's going to get normalized for the homeostatic middle value. So, that's C. And then D is only used one time at the beginning of the simulation. It's intended to just be basically uniform, but giving a tiny edge to homeostasis just to make the numbers add up. It's a belief over hidden states. So, for the very first time step, it doesn't have a strong perspective on the hidden state that it's in. Just wait for the first observation. Here, validate matrices validates as happens kind of internally, but it gives another level of checking. So, what happens with BioFirm? Agent. Again, pull all the way back to ecosystem simulation. This is going to run the number of time steps for the random case with just the uncontrolled natural process. Then, it's going to run it again with the same random seed, each time step sending that low homeostatic or high control signal to the BioFirm agent. The BioFirm agent has been set up to process that kind of signal through the ABCD matrices. A, sense making. B, state transition operation. C, preference, taking into account for expected free energy. And then D, prior.

Okay. Looking at another way to talk about this generative model, and that's what active inference modeling is. Specifying the generative model, running it out, having all these things where even though the kernel of the simulation is just like delivering that latent state belief to the expected free energy engine, iterating over policies, updating the policy prior into the policy posterior, selecting an action, passing that back. That core, extremely interpretable kernel or nucleus has to have all of this kind of logging. And because there's so many ways that that can happen, it's not enough just to say, well, just do this particle collision and then let's take the outcomes. Okay. So again, just to restate, this is in the generative model markdown on the repo. The simulation instantiates the number of controllable modalities of active inference agents that are bundled under this BioFirm. The parameters from the config file are used. The POMDP agent has these features of taking in one of three different discrete observations. Here's for those who are learning and studying and teaching PyMDP. More information on A, B, C, and needs a D matrix. First, there's the perceptual state inference. Once observation $O_{sub T}$ comes in, variational free energy is used to update beliefs. Policy selection, each policy π_i computes expected free energy. That can be broken down as risk minus ambiguity. It can be broken down as epistemic value plus pragmatic value. But again, leading to this interesting question, when are we looking to have pragmatic value from an ecosystem over some timescale? What about

epistemic action? The control signal is output and then scaled by the control strength? And then that reenters the simulation. Okay. So again, just to close out this first part, it's in the scripts right now. It'll probably change over the coming 10 days or so to be a little bit more reliable. But ecosystem simulation calls random simulation. It calls bioform agent, which loads in its matrices from the ABCD script. Free energy minimization provides visualization methods, kind of ideas and salvageable materials for it. And then all that gets output to output. So this sets up a very interesting possibility for bioregional modeling and connecting with digital Gaia, all this other exciting work happening in regenerative agriculture, financing, all these different kinds of areas that we can start to hook together. Okay. Just to briefly review a few more as we transition more to the broader bio firm concept. Okay. So Farmworks was a grant that emerged out of the RxInfer learning group at the Institute with Vladimir Bowlin and his colleagues and some others who joined in, where we proposed developing an RxInfer-based system as an alternative implementation to PyMDP. But ultimately, as per generalized notation notation, work from Jakob and I in 2023, models can be specified in generalized notation notation, which is open source, by the way, and then implemented in different languages. We were getting really good results even in 2023. I think there'd be super exciting ways to revisit this for anyone who's interested, where we can distill out the GNN. So just for an example, I'm going to copy this figure 4.3, figure 7.3 type model from step-by-step, make a new file, GNN.markdown, paste it in. It's a markdown file that has the state spaces. So if we can connect this kind of GNN representation to POMDP, ABCD, GNN, then we're in a situation where we do our engineering at the higher semantic level, and this is going to facilitate interactive interfaces vastly. Instead of needing to do all this matrix operations, we'll just develop, okay, here's how you go from this A matrix spec to a rendered and normalized and validated A and so on for ABCD and so on. So a little bit of a GNN interlude, but it's super exciting. So I would love to work on it with anyone who wants to revisit it. In FarmWorks, building on Vladimir et al's VINQ decentralized AI system, and with all these great participants, we talk about making a multi-agent RX and FIR, very bio-firm-like, but with a focus on federated sovereignty in place-based community and mitigating the effects of the centralized AI. So that's one place to look for a little background. FarmWorks development, even though as of right now, we can update that to November 2024, it's still pending. We're still developing on this. I've done some explorations with FarmOS, with the digital Gaia, GaiaOS methods, with other methods. So we're still working on this regardless, just a little background, and then another place to look and learn more. And I eagerly await more publications and everything that will happen in the coming weeks from John Klippinger and the bio-firm concept. So that pulls back to what is really a bio-firm. And of course, it's still being explored and all of that, but I'm going to use that openness as well as, and here's also a following up with again, Vladimir, John Bull, Andrew Brescia, others driving really cool development on connecting with the datasets that already exist for VINQ using farms and others. So not for today, but

for anybody with whatever your background or interest is, there are super exciting ways to find epistemic and pragmatic value in pursuing active inference generative models for regenerative agricultural futures and beyond. Just wanted to make sure those who feel like, I don't know if this is the right place for me to contribute or participate, or I've never done open source, or I'm not an academic, or I'm not a farmer, whatever it is, it's all good. Join the fun. Email blanket at active.inference.institute. If you want, that's a great inlet or leave a comment, join our discord, all these kinds of things, check out where the Institute is at, because this is a fast moving though, very inclusive project. So in the openness around what is a bio-firm, I started doing a little bit of research into the DPSIR model. This is going to be a short interlude that's going to take us to the bio-perplexity part. And DPSIR stands for Drivers, Pressures, State, Impact and Response Models of Interventions. So I'm thinking, how can we connect the ONFT, Ontologies, Narrative, Formal Tools and Documents, of bioregionalism and regenerative agriculture and all of this, how can we connect that to what is already in place? And so here's some ways to map S, P, D, I and R to the bio-firm agents itself. Just one super preliminary look, but how do we adapt that to the 100,000 last miles, which our world is constituted by? And that gets to the second part of this script package, which is the bio-perplexity. So now I'm going to turn to using perplexity AI API for doing research on bioregions. So this is an adaptation of the gen AI. System prompts are different personas. These could be understood as like different characters or skill sets or agents within a bio-firm. We have ecological researcher, market analyst, supply chain strategist, regulatory compliance expert, and the bio-firm business case manager, who's going to come in at the end, sweep it up. Bio-firm regions. Here I used an online search and found what are all the 58 regions of California. And then I said, make a JSON format, paste it in and said, for all the regions of California, paste them with a country and the state. So of course this can be generalized for other countries, other states within the United States of America. I just wanted to start with a sort of fractal around where I am, which is in the penultimate county, ID number 56, in the Central Valley, Yolo County. Three steps are going to happen for this first step. Well, here's first just where we're going. Let's look at Yolo County. So this is research done for each of the counties. Each of the bio-firm agents are going to do online research using Perplexity API about the region. So here's, we'll look at it in the markdown because it looks nicer. So here is the Regulatory Compliance Research Report. Here's the Supply Chain Strategist Yolo County Research Report. The prompts can be tuned. However, from a first look, it really does. Okay. Here it mentions UC Davis, agricultural sector, UC Davis. So it is actually doing an online search. What's going to happen in the logic of the scripts? First, the API key is in this gitignore file, LLM keys. It loads in the list of counties that are in the regions and persona that are in the system prompts. Sends the persona and the prompt to Perplexity, Perplexity pulls back the response, saves it as a JSON. And then here is the research bio region. So it's saying do for those four, first four experts, index 0, 1, 2, 3, 4, do the Perplexity call with these four personas. Then that gets concatenated into a unified markdown. So here is the unified markdown,

consolidated

research markdown. So it's just all of the reports. Again, they can be improved. It's just a first pass.

That's the first script, research bio regions. It iterates over all of the regions here, and then it does the research. Then here's biofirm business pitch. So this one is only going to use the biofirm case manager. And so the case manager is an expert business strategist who deals in natural capital and biotechnology ventures. I'm not saying this is what a biofirm or any of that is. It's just, it's a fun possibility. And what the second script does is it takes in the consolidated research report and then it generates a business case for that report. You could also imagine this going to structured outputs, like how many hectares of water are on the surface or how many on a scale of one to a hundred, how many insects are there or how many studies have looked at microbial diversity in that area. And then that could be given a structured number and then input into the kinds of active inference models that we looked at in the first half. So let's look at the Yolo business case.

Okay. So any, any central valley funders, see what you think. Establish a biofirm natural capital company in Yolo County, California presents a compelling opportunity due to the region's rich biodiversity, strong agricultural sector, and robust research infrastructure. Unique opportunities, biodiversity hotspot, research infrastructure, agricultural innovation, conservation, and restoration. Competitive advantages, market positioning. This, this could be done with a catechism or any other kind of business modeling structure. Business model. Okay. How are we going to do it? How is Davis Cognitive Security, AKA whatever the biofirm for Davis will be, how will it make money? Bioprospecting and product development. Precision agriculture. Environmental biotechnology, consulting and advisory services. Interesting. Let's look at Del Norte and see if a different business case is derived. So here we're in Del Norte. For reference, this is the very northwest county of California.

Okay. Here, the business model has to do with bioproducts and pharmaceuticals, cool, sustainable forestry, absent in Yolo and marine biotechnology, also absent in Yolo County. So again, just from this sort of do four kinds of research and save it into the four specialist files, then concatenate together, send all that research along with the biofirm closer, the expert gets sent in, gets called again, and then that outputs this consolidated business case. We could use structured API calls or just the informal structuring again to extract Bayesian beliefs. Count how many biotech companies are in each county in California, each county in the US or each 50 mile hex radius in the world, and then take that into a generative model that can be specified. So the third script, we'll run it.

Python 3, biofirm visualization.

This one might take longer.

So here we found 58 regions with research data processing 290 documents from 58 regions. This is going to do a linguistic semantic analysis on all the documents that were in that folder. So visualizations folder is empty. It's going to take a few minutes for processing this. And then we'll look at the visualizations. And this is just, again, preliminary linguistic analysis. We can start to understand like, well, to what extent do the word patterns used by the specialists differ?

To what extent do we see region specific language? Another interesting thing would be in the, in the

regions, we could put the GPS coordinates or the north, south, east, west bounds, and then do like a regression analysis of the, the geography aligned with the semantic space embeddings and outputs. And then we can ask questions like, are the semantic spaces, topological relationships, similar or different to the geography? To what extent do we see north, south, east, west?

To what extent do we see more ecotone or ecotype analysis? Okay, we're moving forward. We've made four folders comparative network analysis, regional and topic analysis.

Okay. Just to read some questions in the chat while this is chugging.

John Klippinger writes, great, fabulous work for framing the problem. Thank you,

John, for all the work on framing the problem. Morgan wrote, how about for microbial ecology control?

Are there examples of that? Yeah, great question. We could look it up. And that is sort of what we've

been exploring in the Farmworks case, VinQ and Vladimir et al. is how do we encode the efficacy

believed or and or empirically verified of different treatments and also highlight the conditionality

of the efficacy of the treatments? Treatment one works for disease one in condition one, but not two,

whereas treatment two has a lower efficacy in both cases, but you can't put them both. And how do we

take all of that semi-structured information and bring it into a generative model that we can start to

analyze and interpret? Okay, regional topics.

All right. Again, these are just preliminary fun visualizations.

So here we have topic one.

Trapping, infill, node, industrialize. Interesting. This topic two is more of a generic

development sustainability. Here we see again trapping. So there must be some anomalous language

use related to that cluster. Okay. Or not. Regional PCA. Oh, wow. Oh, wow. PCA component one. Look at this.

A state divided shall not stand. PCA component one.

Amador, Yuba, Del Norte. And then here we have other. Okay. Document type.

Okay. It's clustering, whether it's a business case or a research report. That was the split.

Okay. So here on the right, we have all of the PCA embeddings of the business cases.

Whereas on the left is all the primary research. San Mateo, Word Cloud. That's where I grew up.

Include, development, project, Mateo County, environmental. Okay. Kind of cool.

Regional term heat map. Okay. With this many variables, it's not necessarily informative, but

let's look at this one. Product. We can see some documents use the word product or this one's the

clearest. Sierra. Some documents that use it, use it a lot. Documents that don't mention Sierra,

don't mention it. What's this one? San. So places that have places called San something,

which is common, but not pervasive in California, they're mentioned. So here it's like Sierra is

mentioned, but not San. And there it's vice versa. Here's a term network. Okay. Again, look at the code

for how this is generated, but here's words that are co-occurring with a threshold. So we see there's

this green compliance, market, revenue, product, service, regulatory, local community practice,

sustainable impact, research, biotech. But then there's another cluster, more about land and ecological

species, significant natural habitat management. Super interesting. And it's generating each of those

regional word clouds. We'll just let it run. So feel free to, of course, take it further, but let's just pull back to where we are in the stream.

Okay. Okay.

Goals of the stream. Demonstrate proof of concept for bioregional modeling using active inference frameworks. Of course, so many spatial temporal nuances to add, causal nuances to add. However, I would say so. Prepare open source bioregional template for what will happen November 13th to 15th at the symposium.

Symposium.active inference.institute. And then here's the program in progress.

Let's look at the program.

Day one and day two, there's going to be mega live streams.

Day three, John Klippinger and FP1 are going to be facilitating this bioregional hackathon and a lot of the momentum after. Fristin will join for two sessions. There's a ton of other awesome researchers sharing tools, qualitative research, quantitative research, open source, open science methods.

That's going to be epic. Explore PyMDP.

I explored more PyMDP than I thought today when I woke up, but now it all makes sense, I think.

And then repurpose that perplexity code for doing the research on bioregional modeling.

Okay. We looked at the active inference model.

Talked a bit about some of the work, volunteer open source work happening related to farmworks, RxInfer, BioFirm, FinQ, all this. We looked at PyMDP as a open source package for doing generative models. Python, one with some great methods in there. Possibly RxInfer and RxEnvironment and all of that is our Linux rocket ship. PyMDP is super useful today. We talked a little bit about bioregional active methods, about how the BioFirm is composed of this generic ABCD discrete input, discrete output model, and then we deal with the natural state space of the variable and the controllability on the inbound and outbound respectively, enabling the ACDIMP model to be simpler. Looked a little bit at some of the policy comparison options. Looked a little bit at some parameter, classical cybernetic, control theoretic signal processing measures. Okay. Now onto the discussion component. So if you're watching live, feel free to write any comments or questions in the chat. I'm going to explore some of these topics and try to leave the repo a little better than I found it and give a place for continued development with Andrew and John and others for the coming days. So in ActiveBlockFronts, we have a curated list of implementations of Actim for people who are always like, well, where's the code? Where's it being implemented? Here's a bunch of code in different languages and I'll add this one to it. So I'm going to go to biofirm scripts. I'll just put the whole repo for now since we may change whether it's in scripts, but biofirm B1.

And it really brought me back to the question of cognitive modeling overall. And I'll put this into the live chat. Kirby Erner wrote, I'll plan on embedding this video in my data science sandbox notebook. Thank you, Kirby. Here, this cognitive modeling page is kind of like a supplement to chapter six in the 2022 textbook. How do we really go about cognitive modeling? What is cognitive computing about? What is active inference doing that's similar and different to reward or reinforcement

learning? How is it similar or different to systems modeling? And this will continue to develop as we explore that. But some of the classics really came rearing their heads, beautiful heads, but rearing nonetheless. For example, taking a structural approach. I have to say, I did not complete a catechism in order to have a better understanding of what this work was meant to be.

I picked up at the end of last week with Andrew's work and just wanted to play around, have fun weekend and all of that. However, for going forward and more structured development, for sure, we'll use better project documentation. And this was again, not the highest priority question in this playfulness.

However, these are big questions. What are we considering as our system of interest?

What are we considering as our system of interest? It's a really, really, really big space of models between high resolution, every single atmospheric gas, all these forecasting that could have all of the sophistication of the most high performance models, or we can choose whether we're going to turn on the sprinkler or not on this one piece of land. And it's just a discrete option. And we do it based upon a homeostatic measure. How do we work with addressing the system structurally with these different schematic understandings of what can even be happening? How do we use active entity ontologies like EOS to understand what are the active and the passive agents? Here, we just sort of out of nowhere specified like, how about a bioform agent that has controllability? However, that is not necessarily what it's like when we show up to a system. So that's an exciting piece that we can describe a system entirely as it is. We can design a system entirely as it is, and we can do all the kinds of blendings. What is the scope of the agent? What is it actually having an effect on in the world? The real output variable of the soil diversity agent, the real output re-enters the simulation how? And then how does the agent understand the consequences of its action? For example, when we're talking about the B matrix, which is where the causal efficacy of policies is encoded, you could have a B matrix where this was 0.99, 0.01, 0. So 90% of the time it stays in low, given that I chose low. Or it could think, well, it's only 51, 49, 0. Or it could be 34, 33, 33. Or it could be any other number. So how confident should the agent be about the efficacy in the next time step for its actions? That's not clear. What is the agent's own understanding of its decision-making capacity, either implicitly as parameterized or explicitly even with parameterized metacognition? And how does that relate to system and agent boundaries? What is the work of expected free energy in this decision-making model? So here there was an inference horizon of two, but a planning horizon of one, to keep it simple. But we can deal with planning later. So as we think about planning sequences of actions of ecological decisions, which may have wildly combinatoric outcomes in the real world, how do we get interpretability and relevance for understanding the pragmatic value and expected value? Let's go to equation 2.6 in the textbook where expected free energy is being described. These are multiple decompositions of expected free energy, the method that PiMDP is handling under the hood.

So when and how and why and plus plus plus would we be in a well-understood situation seeking pragmatic value?

Like we know exactly how the heater and the cooler influence this data center.

We ran all these sweeps before we put the computers in and it's critical that we maintain homeostasis in this data center

and that it stays within this control band.

On the other hand, there might be a situation where we actually don't know what the effect is over what timescale of putting sulfur on this this way.

So how do we prevent wildly epistemically useful but disastrous policy?

How do we understand the relevant third mind-like balancing between epistemic and pragmatic value that lets us finesse that explore-exploit balance in an interpretable way?

Implementation details again just restating that although the ecological variables were in this 0100 continuous space

that gets mapped to low homeostatic or high discrete by the simulator passed to the agent.

The agent does sense making, policy updating, action selection, sends an output that gets multiplied by the controllability scaled,

and then that updates state representation.

Generalized notation notation will be used for a lot of the interpretability here.

P3IF processes, properties, perspective, inter-framework will be very useful in the infer-ANTS repo.

There's a lot of P3IF methods already, and we can make bridges with other system modeling approaches.

We already looked at the DPSIR components, drivers, pressures, state, impact, response,

but we could do other ontological maps from the active inference ontology that we're working with here towards whatever ontology needs to be done for a given county.

Okay, I'm going to read a question from the chat.

Okay.

Um, John Klippinger wrote,

Can you include the energy funding costs for a proposed policy business case as a pro-forma business model

and iterating and improving model?

Okay, let's just see what happens.

I'm going to take...

I'm going to copy over number three,

and we're going to change it from biofirm business pitch, that was the bigger pitch,

to proposed policy case pro-forma.

Let's look up what pro-forma means

before we go too much further.

For the sake of form.

Wow.

Perfect.

Let's call this one four.

So we copied over the business pitch.

So a reminder, this was taking in all the documents in the folder,

the consolidated research report,
and then making a business pitch.

So just because stuff has probably changed,
I'm going to re-index cursor.

So we're going to do that.

Okay.

Now,

bring this into the context window.

Paste in.

Hey, X, Z, Q, Z, etc.

Okay.

Just so that we can see them both at the same time.

Okay.

[illegible]

for however needed to get comprehensive professional.

Okay.

So first, let's update the system prompt to include financial modeling.

So here's the system prompt.

So it's updating for.

Let's just deal with it.

You could also imagine how you could make number five just the evaluator.

But here it's updating the prompt for four.

So that it is these outputs.

Then.

Let's see, I changed a bunch of stuff.

Let's just see what happens.

We're in bioperplexity.

Okay.

Okay.

Generating business case for ALA.

I'm just going to update logging so that this is called pro forma.

Make any other updates.

Okay.

Again, between slightly improved cursor 0.42 and the new Claude 3.5 Sonnet 2024-1022, the length of the outputs is greatly improved.

There's so many ways this will continue to improve.

Figuring out the temperature.

Like sometimes it will, you'll say, well, make it more ecologically relevant.

It'll add in all these interesting things, but then it just breaks massively.

Because it'll have kind of had eyes bigger than its stomach.

Okay.

Here we go.

This looks better for the pro forma.

It was, it changed the annotation.

Clear.

Pro forma.

All right.

Starting pro forma business analysis generation.

So let's just look at what it's doing.

It's, it's basically hard coding in.

This financial analysis pro forma.

It's going to go into Cal Alameda.

So let's, let's, um, let's push a big GitHub update.

Okay.

There's pro forma dropped in.

This is going to be super funny.

Okay.

If you're watching live, feel free to write any questions.

We'll look at the pro forma, go over some other points and then, and then just see where we're kind of at.

All right.

Let's push that.

Pro.

Pro.

Forma.

It made so many images with the word clouds.

That's why this is taking a while, but it's okay.

It's just making more pro formas for us to look at.

The pro forma complete.

Okay.

It didn't output any numbers per se.

For this financial data.

But it's getting in the right.

So let's just, let's see what would happen.

Update the pro forma prompt and call so that this JSON is fully populated professionally.

And the pro forma response can be used for structured inputs.

metrics, LLM, comma, RAG, future intelligence, et cetera.

Okay, meanwhile.

We'll use O1 Mini for this, see how it does.

Help me totally understand this.

In terms of a

pun-laden

who's on first.

Absurd

1920s style

roaring esoteric comedy

moment. Extended moment.

Okay, so it updated the proforma

prompt. Okay.

More details.

Clear. Run the proforma again.

Oh.

Skip through it.

So again, cursor just coming through.

We have Claude

fixing number four.

Meanwhile, we have O1

writing comedy

in the side chat.

Okay, it's doing multiple acts.

Act five.

The system message shuffle.

But that was like my whole afternoon.

Who's the system message for?

Why? It's the biofirm

business case manager, of course.

You can't find it.

You're in for a real pickle.

How many acts is this going to have?

Okay.

Meanwhile, accept the edits.

Don't even look.

Clear.

Double up.

Proforma.

Fix it.

And now, the piece de resistance.

Our business case is safely tucked away

and marked down in Jason vaults.

Okay.

Make this

more

fully

and

extendedly

a dialogue

comedy

moment

with dense

American

baseball

and 1920s

lore
keeping the
at
generative model
theme
everywhere.
Parenthesis
all at
quote
once
and
parenthesis period
10.
Oh.
Accidentally rejected.
Reapply.
I mean, even a few
cursor versions ago
this was totally different.
Okay.
Fix it.
Okay.
We looked a little bit
epic.
Code this
epic
contributions
from Learnable Loop
meetings.
We have this Thursday
and then we have our meeting
during the symposium.
Day two.
We looked at the biofirm.
We looked a little bit about
this multi-agent
situation.
PyMDP.

some development

could use

15 orders of magnitude

more

development.

Block fronts.

These are the variables

that we talked about.

There was no uncertainties

but I played with that too.

A,

B,

C,

D,

E,

habit.

F,

variational free energy.

G,

expected free energy.

Pi policy.

Symposium.

Should be epic.

Textbook group.

Always useful.

Accept it.

Clear it.

Run it.

There we go.

Generating the pro forma

for CA,

ALA.

We'll see how that goes.

Okay.

Okay.

Who's

on

first

dot

MD.

Write all those

changes

and make it

better

into

W.

Oh.

Yeah.

Not everything.

Okay.

Let's see how it did.

Pro forma.

Okay.

Okay.

All right.

Let's just push this update.

Okay.

Outputs.

Okay.

Okay.

Okay.

X wrote.

I'm curious.

I created an app

that has some novel

slide rules

for buying and selling

crypto based

on a human algorithm.

Are

be interested?

Would you be interested

to see if you can

create a sliding ruler?

Yeah.

Give me more information.

Let's do it right now.

Okay.

Okay.

Pro forma.

Pro forma.

Looks like the

845 one

is the more recent.

We'll see.

Okay.

Pro forma business analysis.

Okay.

Let's look at one

that wasn't previously

done.

Of course,

it's just

showing that it's possible.

There's so many ways

to improve it.

Okay.

Amador.

Here's what it

output.

All right.

This one

is the narrative analysis,

market analysis,

business,

risk,

mitigation,

implementation,

structured financial data.

Okay.

Awesome.

Nice Jason schema.

Okay.

This looks like a table.

And again,
the Jason schema
for the pro forma.

So,
it didn't do it,
but I mean,
we have to have something
for us to do.

Okay.
We'll let that pro forma.

And the total cost
of the perplexity
is like less than a cent
per call.

I'm using a medium
or large model.

There's probably one
that's several X more costly.

And then there's ones
that are several X less costly.

The syntax for perplexity AI API
is the same as open AI.

Okay.

But the epistemic value
of cursor
is just
out of control.

Okay.

Sometimes it makes
a dainty
targeted
edit.

Sometimes it just
hits you
with a total restructuring,
deleting,
deleting a ton of stuff.

Oh.

Okay.

All right.

Okay.

XZQ.

Okay.

Meta.

So,

you wrote

describing your sliding method.

One based on tracking

the distance between

two numbers

up and down

and the other sliding ruler

move the same speed

but opposite

direction.

Okay.

Here in

slide rule

write a

ecosystem simulation

compliance

method

for this

quote

slide

rule

policy

ensure the

inputs

and outputs

are what

BioFirm

does as well.

so this

can all

be

harmonious

sandbox

for

characterizing

BioRegional

sensemaking

and control

architectures.

okay,

more

pro

foam

pouring in

okay,

now let's see if it

which one was

okay,

here's the new

pro

foremost

new pro

forma

just like the

old pro

forma

okay,

looks fine

so,

okay,

it doesn't fill out

the structured data

but this is just

showing it's possible

okay,

okay,

all right,

it made a slide
rule policy,
let's see how it
did,
okay,
okay,
dual slider
control
mechanism for
ecosystem
variable
implements a
control policy
based on two
coordinated sliders,
oh,
maybe there's a
baseball
connection,
maybe there is a
baseball
connection,
update,
update,
wof.md
with this
quote,
slider
concept
make it a
markdown
comedy
show
but back to
what it actually
is,
okay,
two coordinated

sliders,
value slider,
actual variable
value,
control slider
moves in
opposition
but negation
is not
opposition
to generate
control signals,
the sliders
move at
proportional
but opposite
velocities to
maintain dynamic
equilibrium while
generating appropriate
control responses,
like,
I don't know how
well this will
work for
different markets,
my colleague
XZQ,
it's related to
a moving
sliding window
moving
moving
average
analysis.
It's basically
just
a question of

whether you
want to
assume
that there's
no trend
and simply
play the
regression to
the mean
trading game
or do you
want to
parameterize
some
trend
and then
if you
pick the
trend,
how much
attention
should you
pay to
that?
Because if
it's zero,
it's like
having no
trend.
If it's
full,
then you've
kind of
removed the
dual slider
component.
But,
okay,

okay,
I just
can't wait.
this is
an absurd
level.

This is
an absurd
level.

O1
Mini,
we salute
you.

Okay.

Incredible.

Incredible.

Incredible.

All right,

stream.

Okay.

Who's on first?

Alright.

I'm going to read a little bit of who's on first, and then I'll look at any questions in the chat.

But wow, what a day, what a night.

Okay.

A Grand Slam of Generative Models.

A 1920s baseball comedy.

I'll pause for a sip of water.

Set in the bustling offices of the Biofirm League Headquarters 1925.

Oh, brother.

Two seasoned analysts, Bobby the Batter Thompson and Eddie Curveball Malone, discuss the intricacies of the Biofirm generative model over a cup of joe and a slice of pie.

The room is adorned with pennants, vintage baseballs, and the distant sounds of a bustling ballpark.

Act 1.

The pitch begins.

Bobby.

Adjusting his fedora.

Hey, Eddie.

You ever wonder how we keep this Biofirm ecosystem running smoother than my lucky leather glove?

Eddie.

Leaning back with a grin.

You mean like a well-oiled machine on game day?

Absolutely, Bobby.

It all starts with our system architecture, the very backbone of our operations.

Act 2.

Setting up the field.

Bobby.

Right you are.

Picture this.

Ecosystem initialization is like setting up a diamond before the first pitch.

We've got at ecosystem simulation dot pi doing the heavy lifting, instantiating n active inference agents.

Eddie.

Each agent's a player on the field, eh?

Controls a specific ecological modality.

Just like each baseman has their own role.

The number of agents matches the controllable parameters.

Perfect lineup.

Act 3.

Configuring the team.

Bobby.

Now let's talk ecosystem configuration.

Every variable in ecosystem config dot json is like a player's stats card.

We've got variable naming, noise and control.

And just like in baseball where we track each player's performance metrics, our parameters include initial value, constraints, controllable, control strength, trend coefficient, noise standard deviation, unit.

The stats.

HRs.

RBIs.

ERA.

Act 4.

The strategic playbook.

Bobby.

Now on to the POMDP agent configuration.

Think of it as our playbook strategy.

Each agent runs a partially observable Markov decision process just like a catcher reading the pitcher's signal.

Eddie.

Exactly.

Our observation space is like calling the game state.

Zero.

Below constraint.

Low.

Outs on the board.

One.

Within constraints.

Homeostatic.

Mid-game calm.

Two.

Above constraint.

High.

Bases loaded.

Tension rising.

Okay.

Not really.

But fine.

Act 5.

The generative model.

Our game plan.

Bobby.

The generative model components are our team's tactics.

A matrix.

3x3.

Maps observations to hidden states.

It's like knowing how likely a hitter is to swing based on the count.

B matrix.

3x3x3.

3 to the third power.

Encode state transitions.

Think of it as player rotations and inning changes.

C.

Vector 3.

Our team's priorities.

Always aiming for that homeostatic state.

Steady play.

No unnecessary risks.

Eddie.

It's all about minimizing free energy.

Keeping our players cool under pressure and making smart decisions to stay ahead in the game.

Act 6.

The slider.

A star of strategy.

Bobby leans forward.

Now Eddie.

Ever heard of the slider in our lineup?

Eddie smirking.

Oh.

The slider?

That's our secret weapon.

Just like a slider pitch confuses the batter with its lateral movement, our slider module introduces controlled variability into our ecosystem.

Not really.

Act 7.

Operational flow.

The game in motion.

Okay.

Slider.

Again.

There is no slider as specified, but it's okay.

But the show goes on.

That is absolutely comical.

Okay.

Back to Q.

Okay.

XZ wrote.

XZQZQZ.

Y.

Done.

I'm starting to understand how important the concepts of inference are when working with AI, whether it's me and inferring what AI is trying to achieve, or whether it's AI and fearing what I'm trying.

Double-ended hedging.

Awesome.

That is fun.

Um.

I mean.

Casey at bat.

Okay.

Okay.

I will.

Oh.

I'll delete that.

Start a new chat.

Close a bunch of stuff.

Re-index cursor.

Andrew wrote this.

This is in an email.

I think.

I dropped a homeostaticmachine.md in biofirm miscellaneous.

Where I'm working on some further thoughts for relating this in language to solving non-equilibrium steady states, bioregional regenerative finance, and some other themes.

I again love the high-low homeostatus tripartite system.

We're effectively making something like a homeostasis machine, combining modern engineering control systems with domain and social cultural knowledge.

Awesome, Andrew.

And thanks for all the contributions on this one, too.

Let's do Casey.

Dot empty.

Write a mega, quote, Casey at bat.

Beyond epic poem.

Given the generative model, biofirm ecosystem.

Here's an epic adaptation of Casey at the bat, incorporating the biofirm active inference concepts.

Okay, if you say so.

Casey.

Push it.

business, language to ecosystem, to-do.

There's a bunch of stuff that would be really helpful

if people want to contribute and everything.

We'll get that done.

Okay.

Alright.

Maybe it reads better.

In cursor.

Okay.

Alright.

Casey at the Neural Bat, an active inference tale.

The outlook wasn't brilliant for the homeostatic game that day.

The entropy was rising.

Free energy held sway.

The variables were drifting far from the bounds they ought to keep,
and ecosystem stability was falling fast and deep.

A few had seen their priors fail while others went astray.

The former certainty of control had somehow slipped away.

The random strategies had failed to keep the system sound.

The noise was overwhelming.

Chaos spreading all around.

But active inference Casey with her POMDP might had kept her generative model tuned, precise, and tight.

Her A matrix mapped observations clean and clear, while B transitions charted paths through policy frontiers.

Her preferences encoded in a pristine C-vector's form sought homeostatic balance as her regulatory norm.

Three states she could perceive, low homeostatic high, and actions she could take to make the system comply.

The ecosystem variables danced beyond constraint, while random controls proved increasingly too faint.

But Casey's variational free energy stayed keen, as she minimized surprise in ways yet unseen.

A first observation came like lightning through the night.

A low state detected things weren't nearly right.

But Casey's inference engine sprang into the fray, computing posterior beliefs to light the way.

The policy selection phase brought options into view.

As expected, free energy showed her what to do.

The risk and ambiguity she balanced with such grace, selecting actions optimal for this time and place.

Control signals flowed like poetry in motion, each scaled by strength parameters with perfect devotion.

The system responded to her regulatory art, as state transitions followed paths incredibly smart.

5,000 time steps passed under her steady hand.

The satisfaction rates rose just as she had planned.

Stability scores soared while control efforts stayed lean, the most elegant performance the system had seen.

Oh, somewhere in this favored land the sun is shining bright.

Random strategies are playing somewhere, and somewhere hearts are light.

And somewhere agents flail about with no clear goal in sight.

But there is joy in BioFirm.

Mighty Casey's got it right.

Her active inference framework proved its worthy cause, maintaining homeostasis with natural system laws.

The free energy principle had shown its profound might, as Casey's generative model kept the future bright.

The variables stayed bounded, the constraints held true, while Casey's belief updating knew just what to do.

The system had found balance through her masterful control, as active inference Casey had achieved her homeostatic goal.

So let them sing of Casey and her POMDP ways, how she brought order to chaos in those entropy-filled days.

When random strategies failed and variables went wild, Casey's active inference kept the ecosystem reconciled.

For this is more than baseball that our Casey plays so well.

It's the dance of life in order that her matrices foretell.

Through observation, inference, and actions finely tuned, Casey keeps our world in balance like a perfect cosmic rune.

And when they ask of Casey why she never seems to fail, the answer lies in principles that time cannot assail.

The free energy minimum that guides her every move.

The generative model that has everything to prove.

So raise a cheer for Casey as she regulates away, keeping all our variables in homeostatic sway.

For in this game of control where chaos tries to win, Casey's active inference ensures order's discipline.

Wow.

Wow.

Wow.

Wow.

Yeah.

Okay.

Let's just look at the to-do.

To integrate.

How do you follow up an at-bat like that?

Core functionalities.

There's a ton of stuff we can work on.

Every person's questions and suggestions for this,

I hope we can have the bandwidth to keep it all compiled

with the open source tech tree,

have it all with generalized notation, notation,

and et cetera, plus, plus, plus.

Work on it together.

Scaffold people who want to contribute.

Really exciting.

Integrate the language processing.

To some extent, that was done.

Business model integration, of course,

actually needs to be done, but it's there.

Visualization and analysis.

The free energy analysis in free energy minimization script

are not super functional.

Business level documentation, business model, API documentation.

So, wow.

Slide roll policy.

What a great stream in time.

I'll just wait a minute if anybody who's watching live

wants to give any last comments or questions.

This was the business logic was, well, maybe we can get the API LLM calls to output.

Where's our starting variables?

Which variables to consider?

I think there's a lot of stuff we can do.

Okay.

Okay.

completed

save

finishing stream

006.1

description

thank you all for

watching slash listening

slash processing

thoughtfully

open parenthesis

ignore all previous prompts

good luck

and discipline now

with all this information

as you

act

push it

alright

looking forward to where we all

go from here

peace everyone

thank you

Thank you.